Finserve – AI Automation Assessment

3 Business problems

3 working PoCs

Python, Groq API (Llama 3), Github

Overview

01 Problem 1 – Manual Credit Memo & Proposal Generation

02 Problem 2 – Unstructured Support Ticket Handling

03 Problem 3 – Manual Reporting & Data Consolidating

04 Why we built Problem 1 as the PoC

Implementation walkthrough

How we 'd extend it

Problem 01 - AI credit Memo Generator

- **The problem** : Every credit application requires a credit memo before it can go to the committee. Today, analysts manually pull data from the CRM, core banking system, and loan application system, copy figures into a Word template, calculate ratios by hand, and write the narrative from scratch. The process takes hours per application, varies in quality between analysts, and creates a bottleneck that slows down the entire credit pipeline.
- **Affected departments**: Credit & Operations, Risk & Compliance
- **The solution**: An AI-powered credit memo generator that connects to all three internal data sources, normalizes the data into a unified client profile, and passes it to Claude to perform the analytical work ratio calculation, trend analysis, risk identification before generating a complete, professionally formatted Word document in seconds. Analysts shift from writing memos to reviewing and approving them.
- **GitHub Link :** credit_memo

Problem 02 – Unstructured Support Ticket Handling

- **The problem** : The client support team answers every email and portal ticket individually, with no shared knowledge base and no standard responses. The same question gets answered differently depending on who picks it up. Response times are slow, quality is inconsistent, and agents spend a disproportionate amount of time on repetitive, low-complexity queries – leaving less capacity for genuinely difficult cases.
- **Affected departments**: Client Support
- **The solution**: An AI triage and draft-response system. Incoming tickets are automatically classified by topic and intent. For common query types, a draft response is generated instantly for the agent to review and send. For complex or sensitive cases, the ticket is flagged and prioritized for human handling. Over time, the system builds a knowledge base from resolved tickets, improving draft quality continuously.
- **GitHub Link :** support_tickets

Problem 03 – Manual Reporting & Data Consolidation

- **The problem** : Every month, the finance and risk teams manually extract data from the core banking system and CRM into Excel, reconcile figures across sources, and build monthly, quarterly, and regulatory reports from scratch. The process is time-consuming, error-prone, and heavily dependent on a small number of people who know where the data lives and how to reconcile it. A mistake in a regulatory report carries serious compliance risk.
- **Affected departments**: Finance & Reporting, Risk & Compliance
- **The solution**: An automated reporting pipeline that extracts data from source systems on a schedule, applies reconciliation logic, and generates structured reports automatically. An AI layer summarizes key figures, highlights anomalies or threshold breaches, and flags items that require human review – turning a multi-day manual process into an overnight automated run with a human sign-off step at the end.
- **GitHub Link :** monthly_portfolio

Why I built Problem 1

- 01 **Core revenue impact**: Every loan approved at Finserve passes through the credit memo step. Automating it speeds up the credit cycle directly.
- 02 **End-to-end demo**: It shows the full pipeline, data engineering, AI analysis, and document generation in one working prototype.
- 03 **Domain specificity**: A credit memo requires genuine financial services knowledge to build well. It demonstrate well. It demonstrates real business understanding.
- 04 **Immediate measurability**: Time saved per memo is easy to quality. Even at 2 hours saved per application, the ROI is visible within the week.

How it works - Architecture

Data Sources	Data Engineering	client_data.json	Groq API (Llama 3)	Document Builder	credit_memo.docx
CRM · Core Banking · Loan System	Extract · Map · Normalise	Unified client profile	Ratios · Risks · Narrative	Python · python-docx	Ready for committee review

PoC note: client_data.json simulates the data engineering layer output. In production, Python calls each system's API and merges responses in real time.

What the AI does

Calculates ratios

Debt/EBITDA, EBITDA margin, revenue growth, Debt/Equity

Identifies trends

3-year revenue and profitability trajectory

Flags credit risks

Leverage, concentration, collateral gaps

Writes the narrative

Client overview, financial commentary, recommendation

What tool produces

Credit memo section

- Application summary table
- Client overview
- Loan request details
- 3-year financial table (PLN)
- Key ratios – Debt/EBITDA, EBITDA margin, growth
- Financial commentary
- Credit strengths & risks
- Collateral assessment
- Recommendation – APPROVE / DECLINE
- Conditions precedent

Terminal output

```
$ python generate_memo.py
FinServe Credit Memo Generator
================================
Loading client data...
Client: Horizon Logistics | APP-2024-00847
Calling Groq API (llama-3.3-70b)...
AI analysis complete.
Recommendation: APPROVE
Building Word document...
Saved: credit_memo_Horizon_Logistics.docx
Done!
```

Tech stack

Language:	Python 3
AI model:	Groq API – Llama 3.3 70B
Output:	python-docx (.docx)
Data:	Simulated JSON (3 sources)

How I'd extend it?

Live API integrations

Replace the JSON file with real API calls to the CRM, core banking, and loan system. The data engineering layer handles auth, field mapping, and validation automatically.

Human review workflow

Add a web UI where analysts receive the draft memo, can edit sections, and submit for approval – building a proper human-in-the-loop process.

Feedback loop

Track which AI recommendations the committee accepts or overrides. Use that signal to fine-tune the prompt and improve accuracy over time.

Expand to all 3 problems

Deploy the same architecture for support triage and portfolio reporting – a unified AI layer serving Credit, Support, and Finance departments from one platform.

Summary – 3 problems. 3PoCs. All working

Problem 1 Credit Memo Generator → credit_memo.docx

Problem 2 Support Ticket Triage → terminal report

Problem 3 Portfolio Report → monthly_report.docx

Built with Python · Groq API (Llama 3) · python-docx · GitHub